Grade 12 Prelim Paper 1 August 2024

Memorandum

Question 1

1.1.1. $x(x+4) = 3(x+4)$

$x^2 + 4x = 3x + 12$ ✓A

$x^2 + x - 12 = 0$ ✓CA

$(x+4)(x-3) = 0$

$x = -4$ or $x = 3$ ✓CA both answers

(3)

1.1.2. $6x^2 - 3x = 11$

$6x^2 - 3x - 11 = 0$

$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \dfrac{-(-3) \pm \sqrt{(-3)^2 - 4(6)(-11)}}{2(6)}$ ✓A correct substitution into correct formula

$x = \dfrac{3 \pm \sqrt{273}}{12}$

$x = 1{,}63$ ✓CA or $x = -1{,}13$ ✓CA

-1 Rounding Penalty

(3)

1.1.3 $$\frac{6}{x+3} - \frac{x}{1-x} = \frac{8x-4}{x^2+2x-3}$$

✓A sign change and swap

$$\frac{6}{x+3} + \frac{x}{x-1} = \frac{8x-4}{(x+3)(x-1)}$$ ✓A factorising

$$6(x-1) + x(x+3) = 8x-4$$

$$6x - 6 + x^2 + 3x = 8x - 4$$

$$x^2 + x - 2 = 0$$ ✓CA

$$(x+2)(x-1) = 0$$

$x = -2$ or $x = 1$ ✓CA both Answers

N.A ✓A

(5)

1.1.4 $$-x^2 + 3x > -10$$

$$-x^2 + 3x + 10 > 0$$

$$x^2 - 3x - 10 < 0$$ ✓A

$$(x-5)(x+2) < 0$$

✓C.V. -2

$$-2 < x < 5$$ ✓C.V. 5

✓correct inequality (4)

1.2. $x + y = 3 \quad \ldots\ldots (1)$

$2x^2 - y^2 = 7 \quad \ldots\ldots (2)$

From (1) $y = 3 - x$

into (2): ✓M

$2x^2 - (3-x)^2 = 7$

$2x^2 - (9 - 6x + x^2) = 7$

$2x^2 - 9 + 6x - x^2 = 7$

✓A

$x^2 + 6x - 16 = 0$

$(x+8)(x-2) = 0$

✓CA both x-values

$x = -8 \quad$ or $\quad x = 2$

✓M

$y = 3 - (-8) \quad$ or $\quad y = 3 - 2$

✓CA both y-values

$y = 11 \qquad\qquad y = 1$

OR

From (1) $x = 3 - y$

into (2)

✓M

$2(3-y)^2 - y^2 = 7$

$2(9 - 6y + y^2) - y^2 = 7$

$2y^2 - 12y + 18 - y^2 = 7$

✓A

$y^2 - 12y + 11 = 0$

$(y-11)(y-1) = 0$

✓CA both y-values

$y = 11 \quad$ or $\quad y = 1$

✓M

$x = 3 - 11 \quad$ or $\quad x = 3 - 1$

✓CA both x-values

$x = -8 \qquad\qquad x = 2$

(5)

1.3. $A \times B = (1-x)(1+x)(1+x^2)(1+x^4)\ldots(1+x^{256})$

A ✓ A ✓ ✓A

$= 1 - x^{512}$

(3)

Question 2

2.1. $r = \frac{-9}{27}$

$= -\frac{1}{3}$ ✓A value of r

$-1 < -\frac{1}{3} < 1$ ✓A $-1<r<1$

$\therefore$ The series converges

$S_\infty = \frac{a}{1-r}$ ✓A correct formula

$= \frac{27}{1-(-\frac{1}{3})}$ ✓A value of a

$= \frac{81}{4}$ ✓CA / 20,25

(5)

2.2. $T_n = ar^{n-1} = 126\left(\frac{1}{3}\right)^{n-1} = \frac{14}{729}$ ✓A correct a and r into geometric T_n

✓M equate to last term

$\left(\frac{1}{3}\right)^{n-1} = \frac{1}{6561}$

$\left(\frac{1}{3}\right)^{n-1} = \left(\frac{1}{3}\right)^8$

$n-1 = 8$

$n = 9$ ✓CA

$S_n = \frac{a(r^n-1)}{r-1}$ ✓CA correct substitution into geometric S_n

$S_8 = \frac{126\left(\left(\frac{1}{3}\right)^9 - 1\right)}{\left(\frac{1}{3}\right)-1}$

$= 188,99$ ✓CA

(5)

Question 3

3.1.1. -9 ✓A (1)

3.1.2.

$$\begin{array}{ccccccc} 3 & & -3 & & -7 & & -9 \\ & -6 & & -4 & & -2 & \\ & & 2 & & 2 & & \end{array}$$

$2a = 2$ $\quad$ $3a + b = -6$ $\quad$ $a + b + c = 3$

$a = 1$ ✓A $\quad$ $3(1) + b = -6$ $\quad$ $1 - 9 + c = 3$

$b = -9$ ✓CA $\quad$ $c = 11$ ✓CA

$T_n = n^2 - 9n + 11$

(3)

3.1.3. Consider the sequence of first differences:

$-6 ; -4 ; -2$

correct substitution of a and d into arithmetic T_n ✓

$T_n = a + (n-1)d = -6 + (n-1)(2) = 54$

$2n = 62$

$n = 31$ ✓CA

$\therefore$ The two terms are T_{31} and T_{32}

$T_{31} = (31)^2 - 9(31) + 11$ $\quad$ $T_{32} = (32)^2 - 9(32) + 11$

$= 693$ ✓CA $\quad$ $= 747$ ✓CA

(4)

3.2. $T_5 + T_8 = 50$

$a+4d + a+7d = 50$ ✓A

$2a+11d = 50 \quad \ldots\ldots (1)$

$S_8 = 136$

$\frac{8}{2}[2a+7d] = 136$ ✓A

$2a+7d = 34 \quad \ldots\ldots (2)$

$$\begin{array}{r} 2a+11d = 50 \\ 2a+7d = 34 \\ \hline 4d = 16 \end{array} \quad (1) - (2)$$

$d = 4$ ✓CA value of d

OR

$2a + 11(4) = 50$	$2a + 7(4) = 34$
$2a = 6$	$2a = 6$
$a = 3$	$a = 3$

First three terms $3; 7; 11$ ✓ (4)

3.3. $\sum_{k=2}^{11} 2\sqrt{3^x} = x(\sqrt{3}+1)$

$2\sqrt{3^2} + 2\sqrt{3^3} + 2\sqrt{3^4} + \ldots + 2\sqrt{3^{11}}$

$S_n = \dfrac{a(r^n - 1)}{r-1}$ ✓ Geometric sum

✓ value of a

✓ value of r

$= \dfrac{6\left(\sqrt{3}^{10} - 1\right)}{(\sqrt{3}-1)} = x(\sqrt{3}+1)$

$6(3^5 - 1) = x(\sqrt{3}+1)(\sqrt{3}-1)$ ✓A

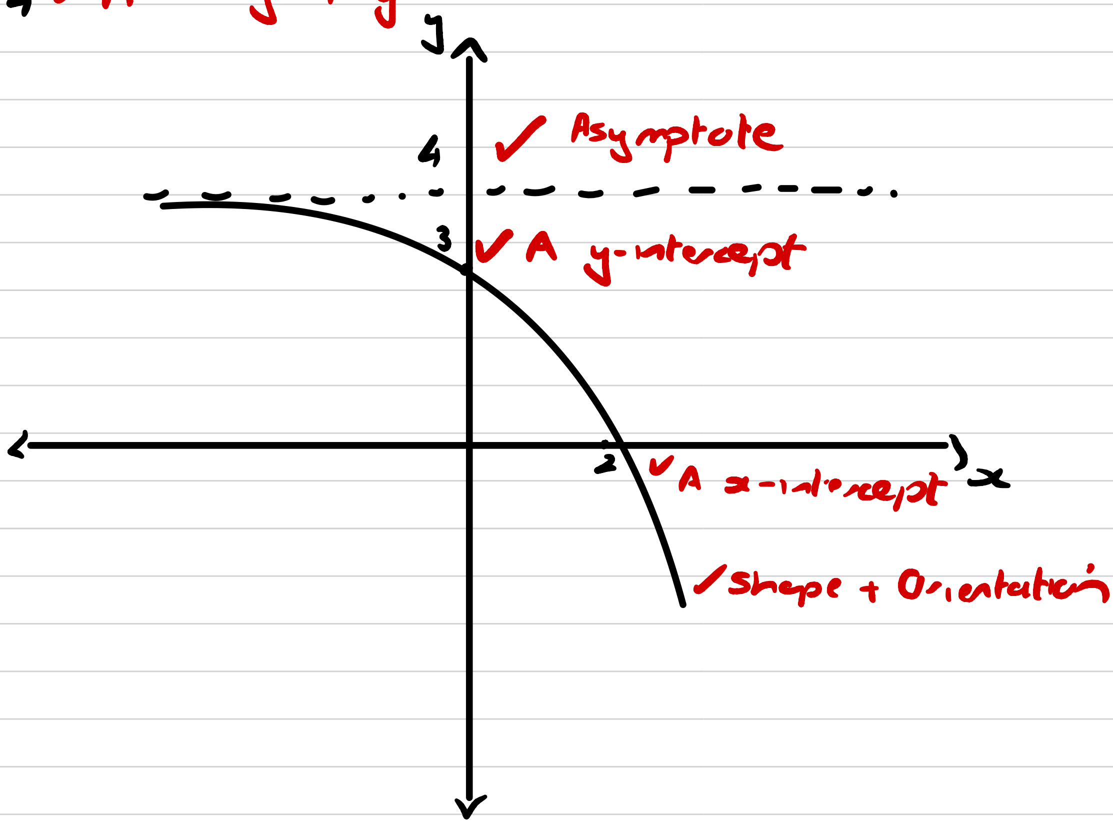
$1452 = 2x$

$726 = x$ ✓CA (5)

Question 4

4.1.1 $y = 4$ ✓A only if $y =$ (1)

4.1.2.



(4)

4.1.3. $a = 2$ ✓A

$b = 1$ ✓A (2)

4.2.1. $x = \frac{-b}{2a}$ ✓M

$x = \frac{-(-4)}{2(-1)}$

$= -2$

$y = -(-2)^2 - 4(-2) - 5$

$y = -1$

✓A ✓CA

$B(-2; -1)$ (3)

OR

$f'(x) = -2x - 4 = 0$ ✓M

$x = -2$

$y = -(-2)^2 - 4(-2) - 5$

$y = -1$

✓A ✓CA

$B(-2; -1)$ (3)

OR

$f(x) = -x^2 - 4x - 5$

$= -(x^2 + 4x + 5)$

$= -(x^2 + 4x + 4 - 4 + 5)$

$= -[(x+2)^2 + 1]$ ✓M

$= -(x+2)^2 - 1$

✓A ✓CA

$B(-2; -1)$ (3)

4.2.2 $x \neq -2$; $x \in \mathbb{R}$ ✓A must have both (1)

4.2.3.

$$y - y_1 = m(x - x_1)$$

$$y + 1 = -1(x + 2)$$

$$y + 1 = -x - 2$$

✓A ✓CA From coordinates of B

$$y = -x - 3$$ (2)

4.2.4

$$f\left(-\tfrac{1}{2}\right) = -\left(-\tfrac{1}{2}\right)^2 - 4\left(-\tfrac{1}{2}\right) - 5$$

$$= -\frac{13}{4}$$ ✓A

$$f'(x) = -2x - 4$$ ✓A

$$f'\left(-\tfrac{1}{2}\right) = -2\left(-\tfrac{1}{2}\right) - 4$$

$$= -3$$ ✓CA

$$y - y_1 = m(x - x_1)$$

$$y + \frac{13}{4} = -3\left(x + \frac{1}{2}\right)$$

$$y + \frac{13}{4} = -3x - \frac{3}{2}$$

$$y = -3x \quad \frac{19}{4}$$ ✓CA

(4)

4.2.5.

$$y = \frac{a}{x+p} + q$$

$$y = \frac{a}{x+2} - 1 \quad \checkmark A$$

sub $(-4;0)$

$$0 = \frac{a}{-4+2} - 1$$

$$1 = \frac{a}{-2}$$

$$-2 = a \quad \checkmark CA$$

$$\therefore \quad y = \frac{-2}{x+2} - 1$$

$$y+1 = \frac{-2}{x+2}$$

$$(y+1)(x+2) = -2$$

$p = 2$ ✓CA ; $q = 1$ ✓CA ; $t = 1$ ✓CA (5)

4.2.6. $-4 \leq x < -2$ ✓A ✓A or $x > -2$ ✓ (3)

Question 5

5.1. $y = -2\sqrt{x}$; $x \geq 0$ and $y \leq 0$

Let $x = y$ ✓M swap x and y

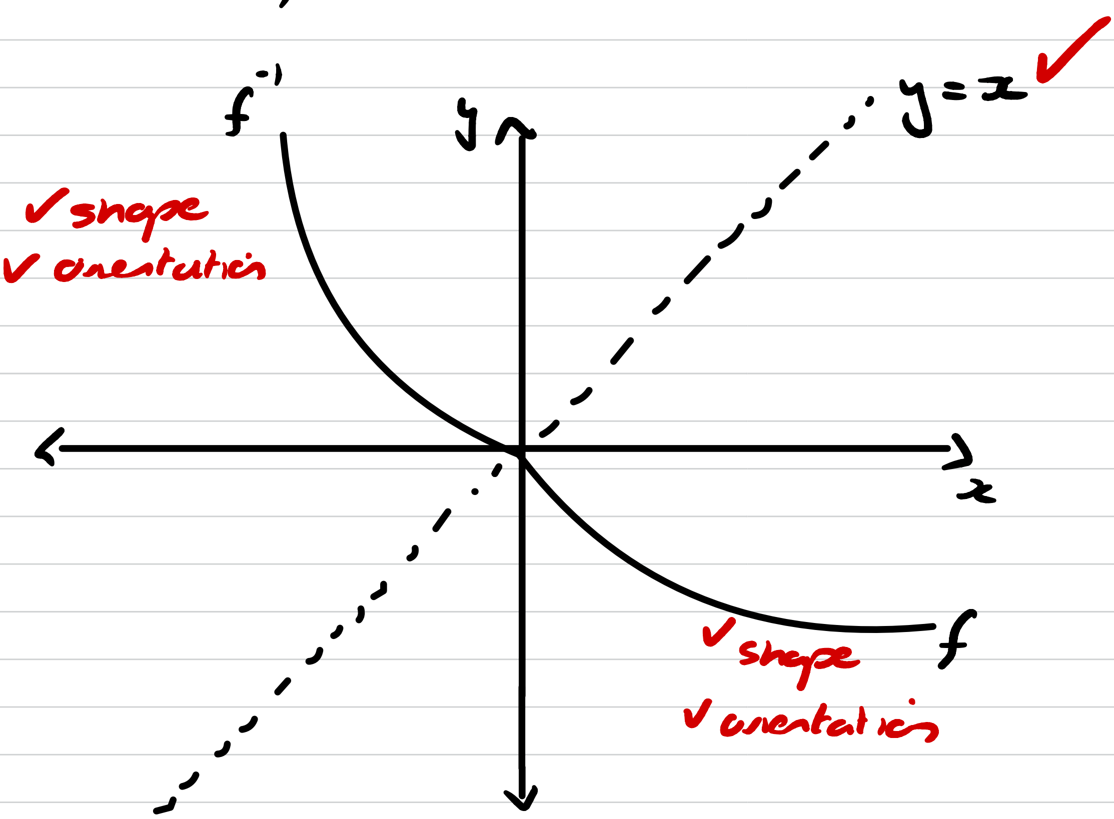
$x = -2\sqrt{y}$; $x \geq 0$ and $y \leq 0$

$\frac{x}{-2} = \sqrt{y}$; $y \leq 0$ and x 0

$y = \frac{x^2}{4}$; $x \geq 0$ ✓A both equation and restriction

(2)

5.2.



(5)

5.3 $(0;0)$ ✓A ✓A

(2)

Question 6

6.1. $1 + i_{eff} = \left(1 + \frac{i_{nom}}{m}\right)^m$ ✓A formula

$1 + i_{eff} = \left(1 + \frac{0,1235}{4}\right)^4$ ✓substitution

$i_{eff} = \left(1 + \frac{0,1235}{4}\right)^4 - 1$

$i_{eff} = 0,1293$ ✓CA only if correct formula is used.

$\therefore$ Effective rate is 12,93 % (3)

6.2. $A = P(1+i)^n$

$17,68 = 11,43(1 + 0,056)^n$ ✓A correct substitution into correct formula

$\frac{1768}{1143} = \left(\frac{132}{125}\right)^n$

$n = \log_{\frac{132}{125}} \frac{1768}{1143}$ ✓M log

$n = 8,01$ years ✓CA answer (3)

6.3.1. $P = x \left[\frac{1-(1+i)^{-n}}{i}\right]$ ✓M correct formula

$120\,000 = x \left[\frac{1 - \left(1 + \frac{0,0875}{12}\right)^{-48}}{\frac{0,0875}{12}}\right]$ ✓A substitution

$x = R2971,98$ ✓CA

(3)

6.3.2. $OB_8 = P(1+i)^n - \frac{x[(1+i)^n - 1]}{i}$

$= 120\,000\left(1+\frac{0{,}0875}{12}\right)^8 - 2971{,}98\frac{\left[\left(1+\frac{0{,}0875}{12}\right)^8 - 1\right]}{\frac{0{,}0875}{12}}$ ✓CA

$= R102\,789{,}73$ ✓CA

$A = P(1+i)^n$

$= 102\,789{,}73\left(1+\frac{0{,}0875}{12}\right)^3$

$= R105\,054{,}69$ ✓CA

$P = \frac{x[1-(1+i)^{-n}]}{i}$

A value of n ✓

$105\,054{,}69 = \frac{x\left[1-\left(1+\frac{0{,}0875}{12}\right)^{-37}\right]}{\frac{0{,}0875}{12}}$

$x = R\,3249{,}81$ ✓CA (5)

Question 7

7.1. $f(x) = -x^3$

$f(x+h) = -(x+h)^3$

$= -(x+h)(x+h)^2$

$= -(x+h)(x^2+2xh+h^2)$

$= -(x^3+2x^2h+xh^2+x^2h+2xh^2+h^3)$

$= -(x^3+3x^2h+3xh^2+h^3)$

$= -x^3-3x^2h-3xh^2-h^3$ ✓A value of $f(x+h)$

$f'(x) = \lim_{h\to0} \frac{f(x+h)-f(x)}{h}$

$= \lim_{h\to0} \frac{-x^3-3x^2h-3xh^2-h^3-(-x^3)}{h}$ ✓correct substitution

$= \lim_{h\to0} \frac{-x^3-3x^2h-3xh^2-h^3+x^3}{h}$

$= \lim_{h\to0} \frac{-3x^2h-3xh^2-h^3}{h}$

$= \lim_{h\to0} \frac{h(-3x^2-3xh-h^2)}{h}$ ✓Factorise

$= \lim_{h\to0} (-3x^2-3xh-h^2)$

$= -3x^2-3x(0)-(0)^2$ ✓Apply limit

$= -3x^2$ ✓Answer (Answer only 0/5) (5)

-1 Notation penalty for incorrect lim

eg lim =; not writing lim in correct steps.

7.2.1. $y = \frac{3}{x^2} - 10\sqrt[5]{x}$

$y = 3x^{-2} - 10x^{\frac{1}{5}}$ ✓A

$\frac{dy}{dx} = -6x^{-3} - 2x^{-4/5}$

✓CA ✓CA (3)

7.2.2. $y = \frac{x^3+1}{x+1}$

$y = \frac{(x+1)(x^2-x+1)}{x+1}$ ✓A

$y = x^2 - x + 1$

$\frac{dy}{dx} = 2x - 1$

✓CA ✓CA (3)

Question 8

8.1. 3 ✓A (1)

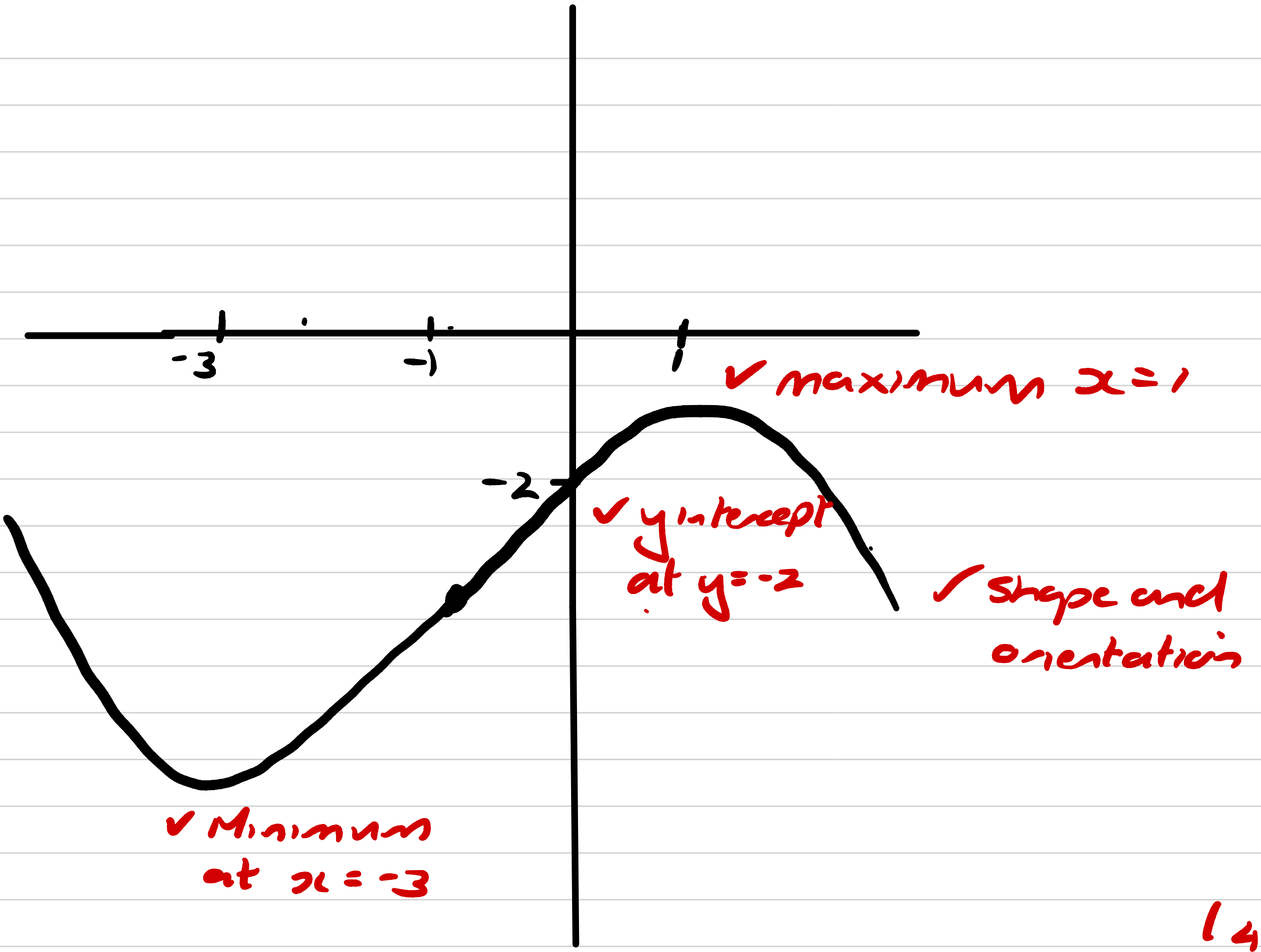
8.2. $x = -3$ ✓A and $x = 1$ ✓A (2)

8.3. 3 ✓ (1)

8.4.1. $-3 \leq x \leq 1$ ✓correct values ✓correct inequality (2)

8.4.2. $x > -1$ ✓value ✓inequality (2)

8.5


-3
-1
1
-2
✓ maximum x=1
✓ y intercept at y=-2
✓ shape and orientation
✓ Minimum at x=-3

(4)

8.6. $f(x) = ax^3 + bx^2 + cx - 2$

$f'(x) = 3ax^2 + 2bx + c$ ✓ Derivative in terms of a; b; c

Find equation $f'(x)$:

$y = a(x - x_1)(x - x_2)$

$y = a(x+3)(x-1)$ ✓M

Sub $(0;3)$

$3 = a(0+3)(0-1)$ ✓ sub

$a = -1$

$y = -(x+3)(x-1)$

$y = -(x^2+2x-3)$

$f'(x) = -x^2 - 2x + 3$ ✓ Derivative

$\therefore$ $3a = -1$ $\quad 2b = -2 \quad c = 3$

$a = -\frac{1}{3} \quad b = -1$

$f(x) = -\frac{1}{3}x^3 - x^2 + 3x - 2$ ✓ CA Equation

(5)

Question 9

9.1.

$$x^2 + r^2 = 9^2$$ ✓ pythag

$$r^2 = 81 - x^2$$

$$V = \pi r^2 h$$ ✓

$$= 2\pi x(81 - x^2)$$ ✓

$$= 162\pi x - 2\pi x^3$$ ✓A derivative (3)

9.2. $$V' = 162\pi - 6\pi x^2 = 0$$ ✓M =0

$$x^2 = 27$$

$$x = 3\sqrt{3}$$ ✓CA x-value

$$\therefore V = 162\pi(3\sqrt{3}) - 2\pi(3\sqrt{3})^3$$ ✓M sub

$$= 1763{,}01\ cm^3$$ ✓CA answer

−1 Unit penalty (5)

Question 10

10.1. $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

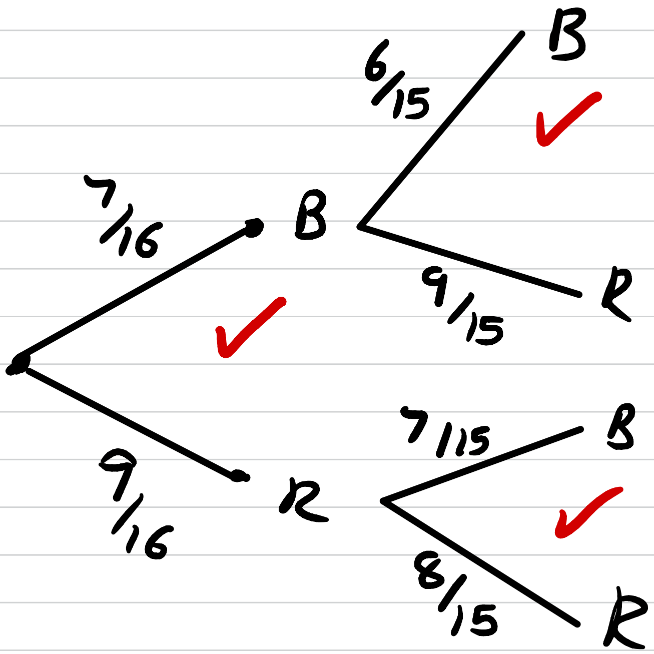
✓ $P(A)$ ✓ $P(A).P(B)$

$= 0{,}68 + 0{,}17 - 0{,}68 \times 0{,}17$

$= 0{,}73$ ✓ CA answer

(3)

10.2.1.



BB
BR
BR
RR

✓ outcomes

(4)

10.2.2. $\frac{9}{16} + \frac{8}{15} = \frac{3}{10}$ ✓CA (1)

10.2.3. $\frac{7}{16} \times \frac{9}{15} + \frac{9}{16} \times \frac{7}{15}$ ✓CA

$= \frac{21}{40}$ ✓CA / $0{,}53$

(2)

10.3.1. $\dfrac{\text{Passwords ending in even}}{\text{Total passwords}}$

$= \dfrac{7^6 \times 3}{7^7}$ ✓A ✓A

$= \dfrac{3}{7}$ ✓CA / 0,43 (3)

10.3.2. $\dfrac{7! - 6!2!}{7!}$ ✓A ✓A

$= \dfrac{5}{7}$ ✓CA / 0,71 (3)